CAMBRIDGE - DATA SCIENCE

Applying Statistics and Core Date Science Techniques in Business

Customer Segmentation with Clustering

Analysis Report

Author: Laura Ortega

Project - June 2026

Table of Contents

Introduction **3**

Methods **3**

Dataset 3

Exploratory Data Analysis 3

Clustering 3

Results and Discussion **4**

EDA 4

Selecting k 5

K-Means Clustering 7

Hierarchical Clustering 8

Dimensionality Reduction 8

Conclusion **10**

References **10**

Introduction

In an increasingly competitive global e-commerce landscape, understanding customer behaviour is essential for effective marketing. Customer segmentation (the process of grouping customers based on shared characteristics) enables businesses to move away from one-size-fits-all strategies and towards target data-driven marketing.

This project applies unsupervised machine learning clustering techniques to a real world transnational e-commerce dataset to identify meaningful customer segments and inform strategic marketing decisions. The goal is to develop a robust segmentation that allows the business to allocate resources efficiently, improve customer retention and maximise lifetime value.

Methods

Dataset

The Dataset (SAS, 2024) contains 951,668 rows, each representing a product ordered by a customer, with 20 features covering order details, customer demographics, and financial metrics. Data spans from 1st of January 2012 to 30th of December 2016, covering customers from 47 countries across five continents. After removing 21 duplicate rows, the working dataset comprised 951,648 records. Following aggregation, the final dataset comprised 68,300 unique customers.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed on both the raw dataset and the engineered features to understand distributions, identify outliers, and assess inter-feature relationships. Missing values were identified in City (135), Postal_Code (3,716), and State_Province (117,192); as none of these columns are used as clustering features, they were not addressed. Descriptive statistics and boxplots were examined for all 5 engineered features.

Clustering

Five customer-level features were engineered from the raw data as required by the project specification:

- Frequency: total number of orders per customer, indicating purchase regularity.
- Recency: days since the customer's last delivery, relative to 30th December 2016 (fixed reference for reproducibility)
- CLV (Customer Lifetime Value): total revenue generated per customer.
- Avg_Unit_Cost: mean unit cost per customer, reflecting preference for low-cost or premium products.
- Age: derived form Customer_BirthDate.

All five features were standardised using StandardScaler, and scikit-learn Pipeline was implemented to encapsulate the scaling and modelling steps. Three methods were used to determine the optimal number of clusters (k), as summarised in Table 1.

Method	Type	Description	Parameters
Elbow Method	Evaluation	Plots WCSS against k=2-10; the point of deceleration indicates optimal k	K = 2 - 10
Silhouette Score	Evaluation	Measures cohesion vs. separation; range -1 to +1	Sample_size = 10,000
Davies-Bouldin Index	Evaluation	Measures average cluster similarity; range 0 to ∞	—
K - means	Machine Learning	Partitions customers into K clusters by minimising within-cluster sum of squares (WCSS).	K=5, n_init=10, random_state=42
Hiercarchical (Ward)	Machine Learning	Agglomerative clustering using Ward linkage. Dendogram used for independent validation.	Method: “Ward”, sample n=30,000

Table 1: Summary of clustering methods used in this project.

Results and Discussion

EDA

Summary statistics for the five engineered features are presented in Table 2. No missing values were identified in the engineered features.

	Frequency	Recency	CLV	Avg_Unit_cost	Age
Count	68,300	68,300	68,300	68,300	68,300
Mean	13.93	366.40	1,950.15	78.90	43.13
std	11.33	414.29	1,719.92	38.01	17.49
min	1	-20	0.63	0.50	19
25%	6	44	696.00	57.58	28
50%	11	188	1,497.40	73.45	43
75%	19	538	2,709.84	92.52	58
max	121	1,824	18,860.96	1,463.50	74

Table 2: Summary statistics for the five engineered customer features.

The boxplots in Figure 1 confirm right-skewed distributions for Frequency, CLV, and Avg_Unit_Cost, with numerous high-value outliers (a characteristic pattern in retail data). Recency shows a minimum of -20 days, indicating minor data entry inconsistencies that do not affect the clustering analysis. Age follows a broadly uniform distribution between 19 and 74 years.

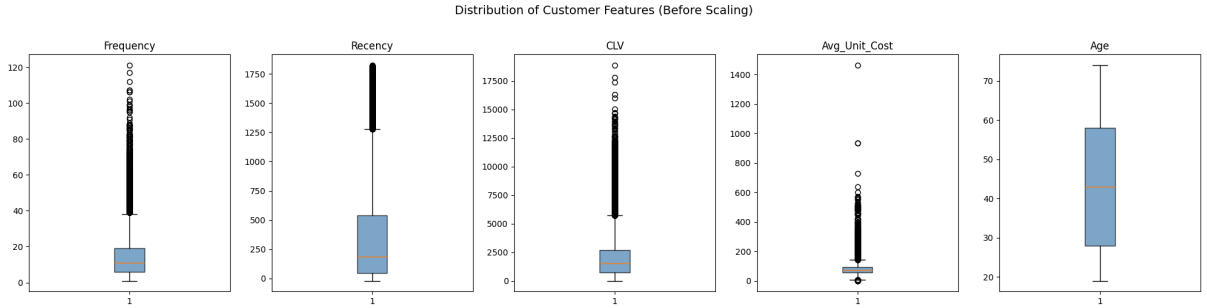


Figure 1: Boxplots of the five engineered customer features before scaling.

Selecting k

Table 3 presents the results of the three k-selection methods across k=2 to k=10. The elbow method plot (Figure 2) shows a marked deceleration in WCSS after k=5. The silhouette score peaks at k=5 (0.2649). The DBI reaches its minimum at k=6 (1.1289), with k=5 scoring near-identically at 1.1335. The convergence of all three methods on k=5 provides strong justification for this choice. Selecting k=5 over k=6 also yields more interpretable and actionable customer segments.

k	WCSS	Silhouette Score	Davies-Bouldin Index	Notes
2	249,552	0.2632	1.3488	
3	204,804	0.2553	1.4303	
4	171,007	0.2519	1.2626	
5	**145,073**	**0.2649**	**1.1335**	**Selected**
6	129,340	0.2470	1.1289	
7	119,689	0.2295	1.1648	
8	110,826	0.2345	1.1948	
9	102,804	0.2238	1.1683	
10	96,932	0.2252	1.1845	

Table 3: k-selection results for the elbow method, silhouette score, and Davies-Bouldin Index.

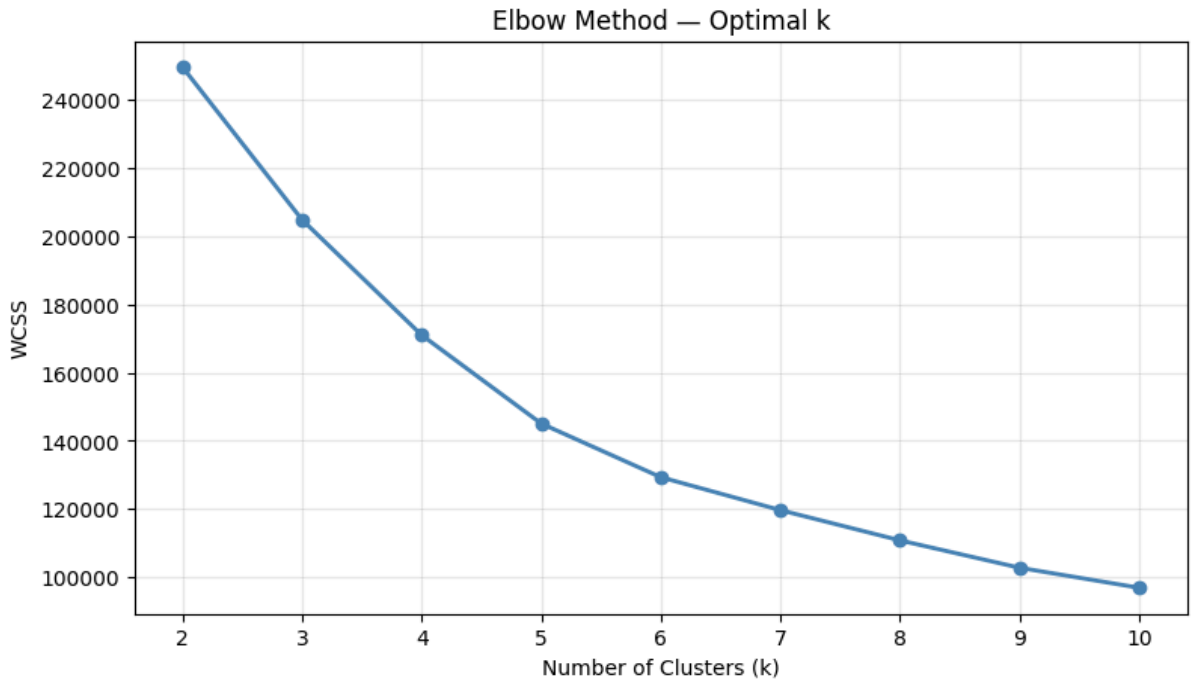


Figure 2: Elbow method plot showing WCSS against number of clusters k=2–10.

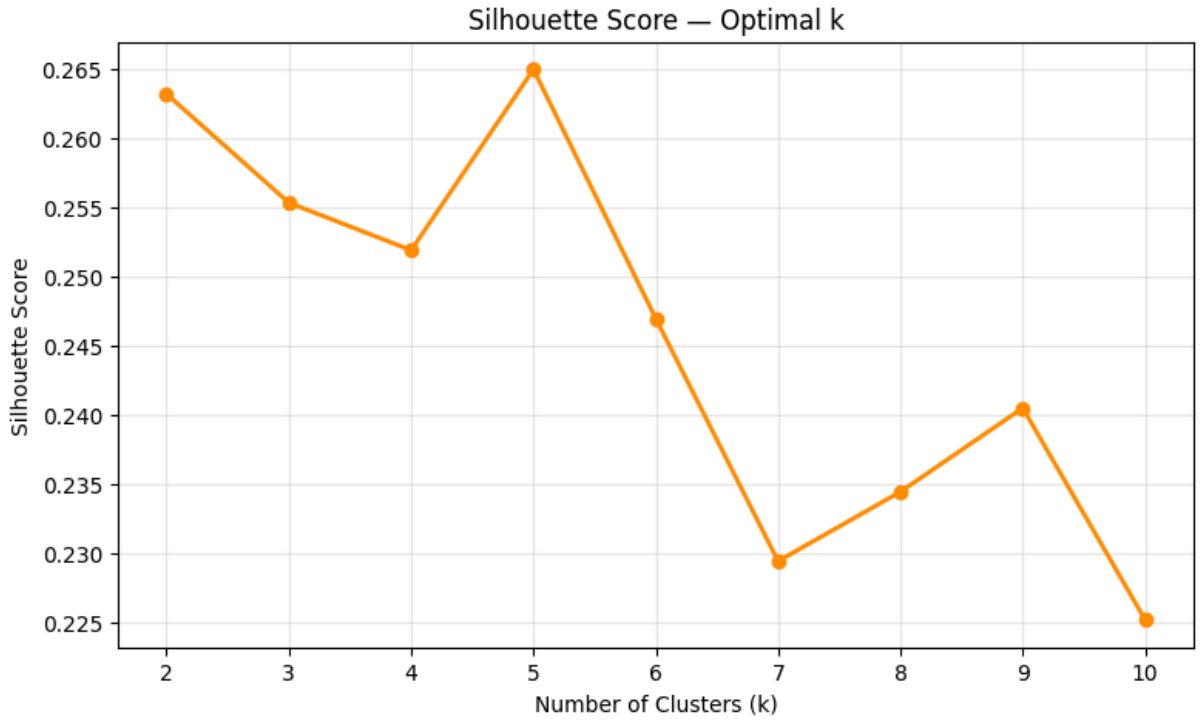


Figure 3: Silhouette score vs k. Peak at k=5 (0.2649).

K-Means Clustering

K-means with k=5 was fitted on the scaled data. The cluster distribution and mean feature values are presented in Table 4, revealing five clearly differentiated customer profiles.

Cluster	Profile	n	Frequency	Recency (days)	CLV	Avg_Unit_cost	Age
0	Older Loyal Customers	20,191	10.99	225	1,411	71.0	60
1	Churned Customers	10,582	5.03	1,116	609	63.69	48
2	Young Active Customers	22,565	12.57	229	1,650	74.13	28
3	VIP/Champion Customers	11,226	33.20	120	4,914	82.87	39
4	Premium Infrequent Buyers	3,736	5.38	573	1,570	181.29	45

Table 4: Cluster distribution and mean feature values for k=5.

Cluster	Profile	Recommended MArketing Strategy
0	Older Loyal Customers	Loyalty rewards and personalised communications to sustain engagement.
1	Churned Customers	Win-back campaigns with targeted discounts.
2	Young Active Customers	Digital and social media marketing to capitalise on growth potential.
3	VIP/Champion Customers	Exclusive benefits, early product access, and premium customer service.
4	Premium Infrequent Buyers	Targeted promotions on premium products to increase purchase frequency.

Table 5: Recommended marketing strategies per segment.

Hierarchical Clustering

Hierarchical clustering with Ward linkage was performed on a 30,000 customer sample. The truncated dendrogram (Figure 4) reveals a clear two-group macro-structure: the orange sub-cluster merges at distance ~130, then joins the green cluster at ~190 to form one macro-group; the red sub-clusters merge at ~160 and form the second macro-group; the two macro-groups merge at the top (~270). The largest distance jump (185 to 270) indicates that the most natural cut yields k=2 clusters. While this confirms the presence of a strong binary structure (broadly active vs. inactive customers), k=5 is retained for greater marketing granularity, as supported by the k-means evaluation metrics.

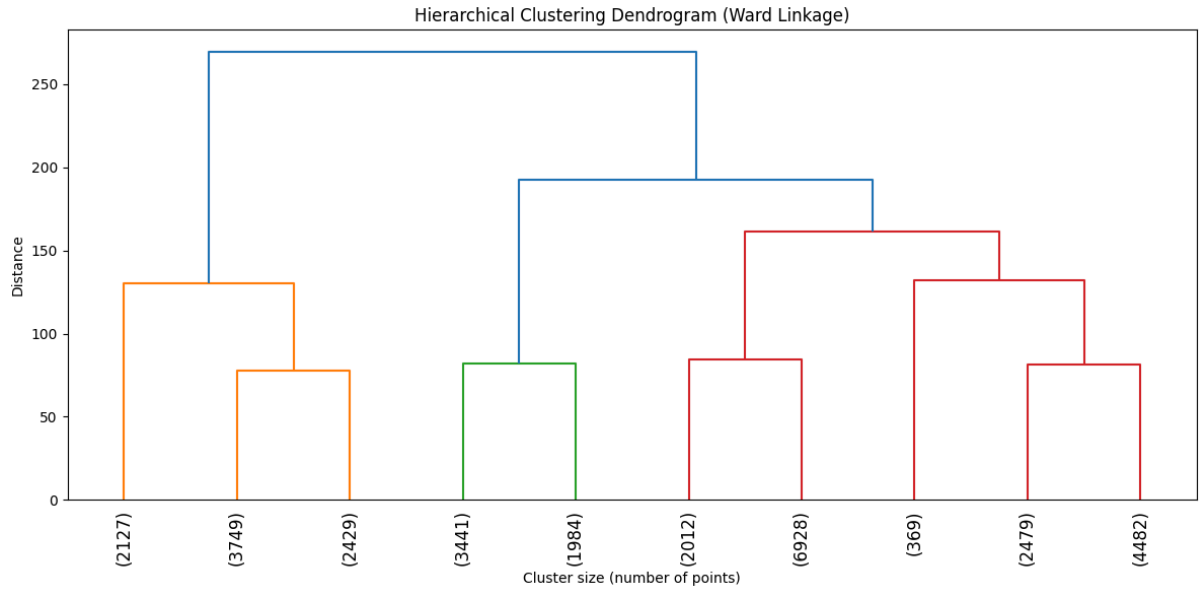


Figure 4: Truncated hierarchical clustering dendrogram (Ward linkage, 30,000-customer sample).

Dimensionality Reduction

PCA reduced the five features to two components capturing 65.25% of total variance (PC1: 44.53%, PC2: 20.73%). While the PCA plot revealed some separation - churned customers (Cluster 1) extending along PC1 - the t-SNE visualisation (Figure 5 and 6, 10,000-customer sample, perplexity=30) provided superior cluster delineation. Young Active Customers (Cluster 2, green) and Older Loyal Customers (Cluster 0, blue) form distinct regions, while Premium Infrequent Buyers (Cluster 4, grey) are visibly separated to the right. The overlap between Cluster 1 and Cluster 3 in the centre is consistent with their similar frequency and unit cost profiles. t-SNE outperformed PCA due to its non-linear nature and ability to preserve local data structure.

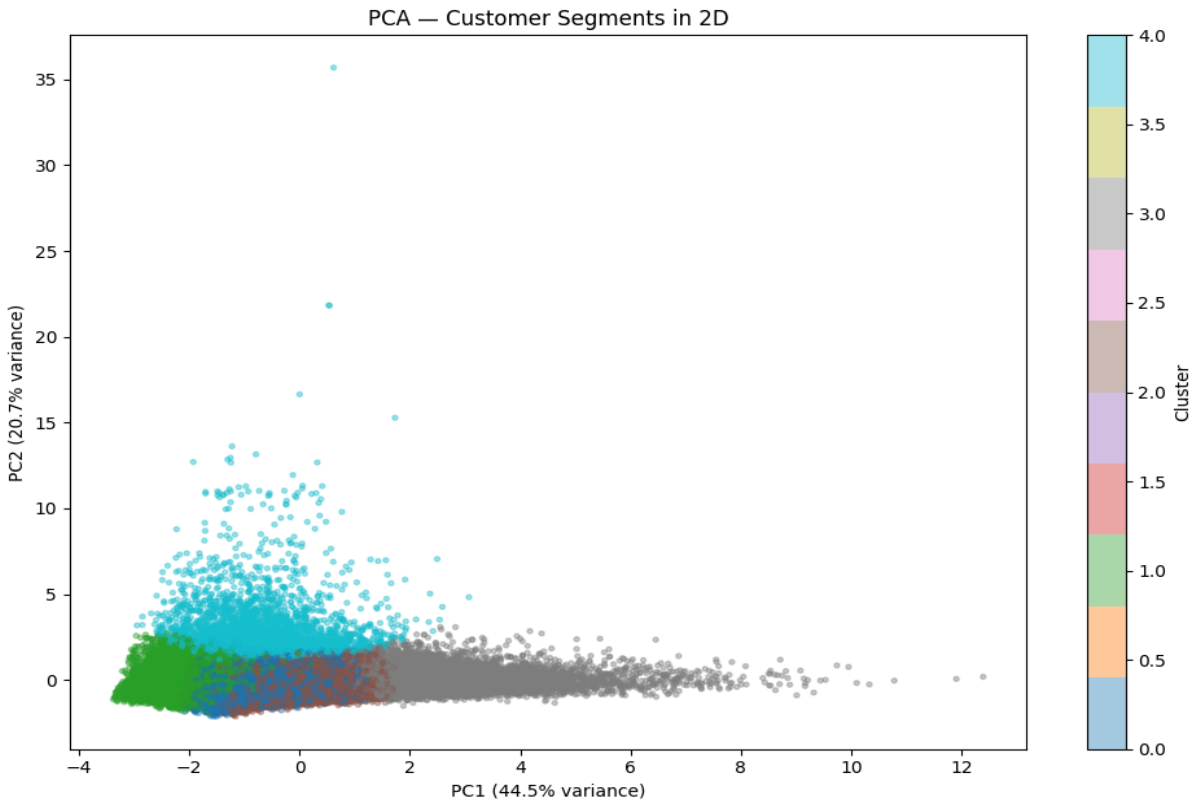


Figure 5: PCA visualisation — 68,300 customers (PC1: 44.53%, PC2: 20.73%).

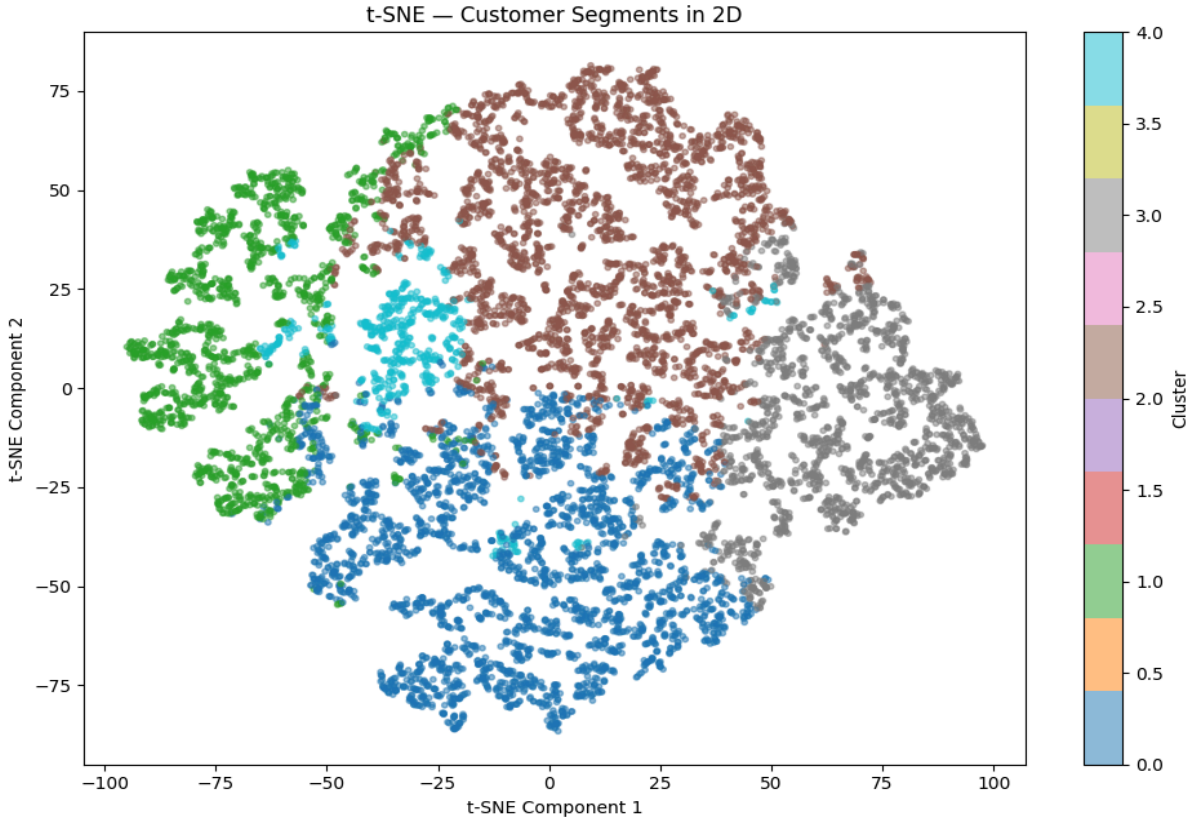


Figure 6: t-SNE visualisation of the five customer clusters (10,000-customer sample, perplexity=30)

Conclusion

This project identified five meaningful customer segments using k-means (k=5), validated by the elbow method, silhouette score, Davies-Bouldin Index, and hierarchical clustering. The segments - Older Loyal, Churned, Young Active, VIP Champion, and Premium Infrequent - exhibit clearly differentiated profiles that directly support targeted marketing allocation. T-SNE outperformed PCA in visualising cluster structure, confirming genuine non-linear patterns in the data. Future work could incorporate product category preferences and geographic features, and explore dynamic segmentation to account for seasonal behavioural shifts.

References

SAS, 2024. CUSTOMERS_CLEAN [Data set]. SAS. Last revised on 15 December 2021. [Accessed 20 February 2024].

Word count: ~780 *(excluding tables and figure captions)*